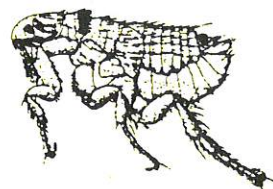


flea NEWS 44



Allen Barnes

Department of Entomology Iowa State University Ames, Iowa 50011-3222

This issue will complete the twelfth year that we have produced **Flea News** here at Iowa State University. During this time we have seen the passing of a shocking number of our colleagues, the publication of many books and research papers on the various aspects of pulicology and a constant change in our readership. The time has come again for us to update our mailing list, particularly in view of the increasing costs of production and postage. Our mailing of **Flea News 43** included 192 copies to recipients in 39 countries.

In order to bring our mailing list up to date, the cover sheet of this issue contains a response for those who wish to continue to receive this newsletter. It should be detached and returned to us no later than the end of November, 1992. In the past we have sent copies to all on our mailing list one issue beyond the deadline. Financial considerations prevent us from continuing this practice, and lack of response will terminate your receipt of this publication after this issue.

MISCELLANEA

In response to my query as to the whereabouts of the flea collection belonging to the late Kôehi Sakaguti, Dr. Bill Jameson questioned some of his Japanese colleagues. He recently contacted us to report that the collection will soon be housed in the Kôbe Museum of Humans and Nature, is to be inventoried, and will then be available for study, either in Kôbe or on a loan basis. Persons interested in this collection should enquire at the following address: Natural History Museum, Administration Office, Hyogo Prefectural Board of Education, Nak Yamate-Dori, 6-1-1 Kôbe 650, JAPAN.

K. S. Larsen (*q. v.*) reports that his laboratory (Statens Skadedyrlaboratorium, Skovbrynet 14, DK - 2800 Lyngby, DENMARK) has *Monopsyllus sciurorum* in culture. They are using the long-tailed field mouse (*Apodemus sylvaticus*) as a laboratory host. Evidently this flea poses a problem to commercial mink farmers where large infestations cause heavy losses from anaemia, growth retardation and death to the kits. Interestingly, the North American squirrel flea, *Orchopeas howardi*, poses a similar threat to commercial mink farmers in Iowa if the ranches are poorly maintained.

JUNE - 1992

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Tel. (515) 294 1815 - Fax. (515) 294 8027 - Bitnet A1.REL@ISUMVS

Miscellanea (Continued)

The following presentations dealing with fleas were delivered at the annual meeting of the Entomological Society of America held in Reno, NV, 8-11 December 1991.

Haematophagous strategies of cat fleas, (*Ctenocephalides felis*). N. C. Hinkle, P. G. Koehler and W. H. Kern, Jr.

Imaginary insects: paradox for the pest control professional. M. F. Potter.

Natural history of cat flea larvae. W. H. Kern, Jr.

Differential activity of cythioate against female and male *Ctenocephalides felis* (Siphonaptera: Pulicidae). M. W. Dryden.

An evaluation of a synergized pyrethrin/(S)-methoprene pet pump spray against infestations of *Ctenocephalides felis* (Bouché) on cats. W. A. Donahue and R. Young.

The following presentation dealing with fleas was delivered at the North Central Branch meeting of the Entomological Society of America held in Kansas City, MO, 15-18 March 1992.

Cat fleas on dairy calves: Severity of infestation and effects on the hosts. A. B. Broce and M. W. Dryden.

John Lane, of the the Department of Medical Parasitology [formerly Entomology] of the London School of Hygiene and Tropical Medicine (University of London) writes that Professor M. G. R. Varma retired in the autumn of 1991 after almost 40 years of service. Although he taught a number of different courses, his main interest in recent years has been ectoparasites, especially ticks.

Mr. Lane also informs us that Dr. Richard P. Lane, formerly Head of the Vector Biology and Transmission Dynamics Unit of the LSHTM, assumed the position of Keeper of Entomology at the British Museum (Natural History), now called the Natural History Museum, in February of this year. Dr. Lane is a dipterist, with primary interest in phlebotomine sandflies (Psychodidae) and their vector capacities.

The following account was contributed by Malcolm Greenwood, Department of Geography, Loughborough University of Technology, Leicestershire, LE11 3TU, U. K.

"Sitting in my office last springtime, I received a phone call, quite out of the blue, from the Drama department of the BBC2 television, asking if I could harness fleas for a 'flea circus'. They were making a film for TV entitled *Flea Bites* (100 mins.), the plot centering around an old man's relationship with a 12 yr old boy to whom he teaches the tricks of the flea circus. The old man, played by one of UK's leading actors (Nigel Hawthorn), portrays a Polish immigrant living in Nottingham and troubled by the memories of losing his son in the Holocaust. The youngster in the film becomes fascinated by the old man's talents and persuades him to pass on his knowledge.

"This led the BBC to me, asking for full-screen closeup shots of fleas laced up to small chariots, to windmills, to perform sword fights and to walk a tightrope. The first problem was a supply of suitable fleas and after an appeal on local radio, we made contact with a lady who runs a hedgehog hospital and whose 'patients' were only too pleased to be de-fleaed. We used then *Archaeopsylla erinacei erinacei* as our 'stars'.

"The last fair-ground flea circus was at Manchester in the 1970's and no 'performers' remain. Anyway, after 3 months of practice using a microscope, 0.01 mm copper wire and a little help from my friends, we managed to satisfy the film makers.

"They were actually very pleased with the results and made it into a film for the large screen which was shown in the London Film Festival in November. The major showing was actually last Sunday 26/1/92, on BBC2 nationwide television.

"All this gave us quite a lot of media coverage leading up to the showing with 'how we did it' type items. Interviews were broadcast on both breakfast national news and local stations, and I enclose just one to illustrate."

The account was accompanied by photocopies of articles in *Radio Times*, *London Evening Standard*, *The Echo* and *Leicester Mercury*. More evidence that the general public still finds these insects fascinating!

☆☆☆☆☆☆☆☆

Allan Marion Barnes

8-December-1924 • 30-April-1992

It is with profound regret that we report the death of Allan M. Barnes, after an extended battle with cancer.

Allan Barnes was born in Stirling City, Butte County, California. He was awarded the B. A. in the Life Sciences with highest honors by San Diego State College, San Diego, California, in 1953. He then took up graduate studies at the University of California, Berkeley, for two years before suspending his studies to become a Vector Control Specialist in the Bureau of Vector Control, California State Health Department, where he remained until 1967. During this period he directed studies on synanthropic fly ecology and control, rodent-ectoparasite ecology and ectoparasite taxonomy and control, and monitored the entomological aspects of rodent-ectoparasite-disease surveillance. He organized the Cooperative Zoonoses Surveillance Network for the state of California. Also during this period he continued his graduate studies and was awarded the PhD. by the University of California in 1963. His dissertation was titled *A revision of the genus Anomiopsyllus (Siphonaptera: Hystrichopsyllidae) with studies on the biology of Anomiopsyllus falsicalifornicus.*

After short-term consultancies with the Ministries of Health in Java and South Vietnam and the United States Advisory Command in Saigon, he accepted appointment as Assistant Chief of the Plague Branch of the Division of Vector-Borne Infectious Diseases, Center for Infectious Diseases, at Fort Collins, Colorado, in 1970. He was promoted to Acting Chief in 1972 and Chief in 1975, a position that he continued to occupy until his death.

During his tenure with the United States Public Health Service, Dr. Barnes served as a consultant and advisor to the Pan-American Health Organization in Venezuela, Brazil, Bolivia, Peru and Ecuador, the World Health Organization in Burma and the People's Republic of China, and the Ministries of Health of Saudi Arabia, Indonesia and British Columbia. He was an Affiliate Professor in both the Entomology Department and the Graduate School of Colorado State University and a member of a number of professional societies. He also authored or co-authored the technical publications listed below.

Although most of Dr. Barnes' publications dealt with matters of Public Health, he authored or coauthored the following flea taxa.

Taxa Described

- 1965. *Anomiopsyllus walkeri* Barnes
- 1965. *Anomiopsyllus princei* = *Anomiopsyllus martini* Holland, 1965.
- 1965. *Anomiopsyllus oaxacae* Barnes
- 1969. *Tunga monositus* Barnes & Radovsky

Selected Bibliography of Allan M. Barnes

The following bibliography is based upon information provided by Dr. Barnes in a communication dated 1988, plus subsequent citations gleaned from the literature. It is not complete, is not represented as such, and does not include presentations undocumented by citation and pagination.

- 1956. Black, R. J. & A. M. Barnes. Effect of earth cover on housefly emergence. *Public Works* 87(3): 109-111.
- 1957. Barnes, A. M. The oriental rat flea in California. *California Vector Views* 4(10): 61-64.
- 1958. Black, R. J. & A. M. Barnes. Effect of earth cover on fly emergence from sanitary landfills. *Public Works* 89(2): 91-94.
- 1959. Barnes, A. M. & B. Keh. The biology and control of lice on man. *California Vector Views* 6(2): 7-11.
- 1960. Barnes, A. M. & L. Kartman. Control of plague vectors on diurnal rodents in the Sierra Nevada of California by use of insecticide bait-boxes. *Journal of Hygiene* 58(3): 347-355.
- 1961. Keh, B. & A. M. Barnes. Fleas as household pests in California. *California Vector Views* 8(11): 55-58.
- 1962. Barnes, A. M. Ectoparasite control in public health. *Vertebrate Pest Control Conference Proceedings* 1: 250-270.
- 1962. Barnes, A. M. Patterns of distribution among western Siphonaptera. *Bulletin of the Entomological Society of America* 8(3): 137. *presentation - no article*
- 1964. Brooks, J. E. & A. M. Barnes. Recognition of household-invading rodents in California. *California Vector Views* 11(12): 65-70.
- 1965. Barnes, A. M. Three new species of the genus *Anomiopsyllus* (Siphonaptera: Hystrihopsyllidae). *Pan-Pacific Entomologist* 41(4): 272-280.
- 1966. Murray, K. F. & A. M. Barnes. Plague in Tahoe Basin. *California Department of Public Health, Special Report*. 9 pp.
- 1969. Murray, K. F. & A. M. Barnes. Distribution and habitat of the woodrat, *Neotoma fuscipes* in northeastern California. *Journal of Mammalogy* 50(1): 43-48.

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- 1969. Barnes, A. M. & F. J. Radovsky. A new *Tunga* (Siphonaptera) from the Nearctic region with description of all stages. *Journal of Medical Entomology* 6(1): 19-36, figs. 1-33.
 - 1969. Barnes, A. M. Fleas: recognition, biology and control. pp. 148-152. In: J. E. Brooks & T. D. Peck (Eds.). *Community pest and related vector control*. Pest Control Operators of California, Inc., Los Angeles.
 - 1969. A. M. Barnes. Ticks: recognition, biology and control. pp. 153-160. In: J. E. Brooks & T. D. Peck (Eds.). *Community pest and related vector control*. Pest Control Operators of California, Inc., Los Angeles.
 - 1971. Hudson, B. W., M. I. Goldenberg, J. D. McCluskie, H. E. Larson, C. D. McGuire, A. M. Barnes & J. D. Poland. Serological and bacteriological investigations of an outbreak of plague in an urban tree squirrel population. *American Journal of Tropical Medicine and Hygiene* 20(2): 255-263.
 - 1972. Barnes, A. M., L. J. Ogden & E. G. Campos. Control of the plague vector, *Opisocrostis hirsutus*, by treatment of prairie dog (*Cynomys ludovicianus*) burrows with 2% carbaryl dust. *Journal of Medical Entomology* 9(4): 330-333.
 - 1972. Brooks, J. E. & A. M. Barnes. An outbreak and decline of Norway rat populations in California rice fields. *California Vector Views* 19(2): 5-14.
 - 1972. Barnes, A. M. & B. W. Hudson. Fleas. pp. 34-38. In: *Vector Control in International Health*. World Health Organization.
 - 1973. Campos, E. G. & A. M. Barnes. Rediscovery of *Nosopsyllus londiniensis* (Rothschild 1903), in San Francisco. *California Vector Views* 19(12): 87.
 - 1973. Poland, J. D., A. M. Barnes & J. J. Herman. Human bubonic plague from exposure to a naturally infected wild carnivore. *American Journal of Epidemiology* 97(5): 332-337.
 - 1974. Barnes, A. M., L. J. Ogden, W. S. Archibald & E. G. Campos. Control of plague vectors on *Peromyscus maniculatus* by use of 2% carbaryl in bait stations. *Journal of Medical Entomology* 11(1): 83-87.
 - 1975. Barnes, A. M. Chapter 8. Fleas. In: D. J. Womeldorf & T. D. Peck (Eds.). *Community pest and related vector control*. 2nd Edition. (Revised). Pest Control Operators of California Publications.
 - 1975. Barnes, A. M. Chapter 16. Ticks. In: D. J. Womeldorf & T. D. Peck (Eds.). *Community pest and related vector control*. 2nd Edition. (Revised). Pest Control Operators of California Publications.
 - 1976. Von Reyn, C. F., A. M. Barnes, N. S. Weber & U. G. Hodgins. Sylvatic plague from exposure to a rabbit: a documented case and a review of rabbit-associated cases in the United States. *American Journal of Epidemiology* 104(1): 81-87.
 - 1976. Prince, F. M., R. B. Eads & A. M. Barnes. *Traubella*, a new genus of flea (Siphonaptera: Ceratophyllidae). *Journal of Medical Entomology* 13(2): 162-168.

1976. Von Reyn, C. F., A. M. Barnes, N. S. Weber, T. Quan & W. D. Dean. Bubonic plague from naturally infected wild coyotes. *American Journal of Tropical Medicine and Hygiene* 25(4): 626-629.
1976. Barnes, A. M. Problems of rodent control in rural tropical areas. *Proceedings of the International Symposium on Arenaviral Infection of Public Health Importance*. *Bulletin of the World Health Organization* 52: 669-676.
1977. Boyer, K. M., R. S. Munford, G. O. Maupin, C. H. Pattison, M. D. Fox, A. M. Barnes & J. E. Maynard. Tick-borne relapsing fever: an interstate outbreak at Grand Canyon National Park. *American Journal of Epidemiology* 105(5): 469-479.
1977. Von Reyn, C. F., N. S. Weber, B. Tempest, A. M. Barnes, J. D. Poland & M. M. Boyce. The 1974 outbreak of bubonic plague in New Mexico: clinical and epidemiological features. *Journal of Infectious Diseases* 136(4): 489-494.
1977. Barnes, A. M., V. J. Tipton & J. A. Wildie. The subfamily Anomiopsyllinae (Hystrihopsyllidae: Siphonaptera). I. A revision of the genus *Anomiopsyllus* Baker. *Great Basin Naturalist* 37(2): 138-206.
1978. Williams, J. E., D. N. Harrison, T. J. Quan, J. L. Mullins, A. M. Barnes & D. C. Cavanaugh. Atypical plague bacilli isolated from rodents, fleas and man. *American Journal of Public Health, Public Health Briefs* 68(3): 262-264.
1978. Barnes, A. M. Rodent control for public health and safety. *Proceedings of the Eighth Vertebrate Pest Control Conference, Sacramento, California*. pp 158-160.
1979. Poland, J. D. & A. M. Barnes. Plague. pp. 515-597. In: J. H. Steele (Ed.). *CRC Handbook Series on Zoonoses. Section A: Bacterial, Rickettsial and Mycotic Diseases*. Vol. I. CRC Press, Boca Raton, Florida.
1979. Barnes, A. M. Human plague in the southwestern United States: sources of infection and prospects for preventive control. *Proceedings of the 34th International Northwestern Conference on Diseases in Nature Communicable to Man*. pp. 11-14.
1979. Edell, T. A., J. K. Emerson, G. O. Maupin, A. M. Barnes & T. M. Vernon. Tick-borne relapsing fever in Colorado. *Journal of the American Medical Association* 241(21): 2279-2282.
1981. Quan, T. J., A. M. Barnes & J. D. Poland. Yersiniosis. Chapter 43. pp. 623-645. In: A. Balows & W. J. Hausler, Eds. *Diagnostic procedures for bacterial, mycotic and parasitic infections*. 6th Edition. American Public Health Association, Washington, D. C.
1979. 1981. Eads, R. B., E. G. Campos & A. M. Barnes. New records for several flea (Siphonaptera) species in the United States, with observations on species parasitizing carnivores in the Rocky Mountain region. *Proceedings of the Entomological Society of Washington* 81(1): 38-42.

1982. Barnes, A. M. Surveillance and control of bubonic plague in the United States. pp. 237-270. In: Symposia of the Zoological Society of London, Number 50.
1982. Barnes, A. M. & G. O. Maupin. Observations on the biting of humans by *Euhoplopsyllus glacialis affinis* (Siphonaptera: Pulicidae) and a review of its plague-transmission potential. *Journal of Medical Entomology* 19(6): 748-749.
1983. Barnes, A. M. & J. D. Poland. Plague in the United States, 1982. *CDC Surveillance Summaries* 32(3): 19-24.
1983. Messick, J. P., G. W. Smith & A. M. Barnes. Serologic testing of badgers to monitor plague in southwestern Idaho. *Journal of Wildlife Diseases* 19(1): 1-6.
1984. Tomich, P. Q., A. M. Barnes, W. S. Devick, H. H. Higa & G. E. Haas. Evidence for the extinction of plague in Hawaii. *American Journal of Epidemiology* 119(2): 261-273.
1984. Weniger, B. G., A. J. Warren, V. Forseth, G. W. Shipps, T. Creelman, J. Gorton & A. M. Barnes. Human bubonic plague transmitted by a domestic cat scratch. *Journal of the American Medical Association* 251(7): 927-928.
1984. Barnes, A. M. & J. D. Poland. Plague in the United States, 1983. *CDC Surveillance Summaries*. 33(1): 15-21.
1984. Quan, T. J., J. D. Poland & A. M. Barnes. Where has pestis gone? Dangerously egregious taxonomic recommendations. *ASM News* 50(5): 193-194.
1984. Smith, C. R., B. C. Nelson & A. M. Barnes. The use of wild carnivore serology in determining patterns of plague activity in rodents in California. *Proceedings of the Eleventh Vertebrate Pest Conference* 11: 71-76.
1985. Eads, R. B., B. C. Nelson, G. O. Maupin & A. M. Barnes. *Orchopeas* fleas (Siphonaptera: Ceratophyllidae) of the western gray squirrel, *Sciurus griseus*. *Journal of Medical Entomology* 22(6): 630-636.
1985. Quan, T. J., A. M. Barnes, L. G. Carter & K. R. Tsuchiya. Experimental plague in rock squirrels, *Spermophilus variegatus* (Erxleben). *Journal of Wildlife Diseases* 21(3): 205-210.
1985. Barnes, A. M., T. J. Quan & J. D. Poland. Plague in the United States, 1984. *CDC Surveillance Summaries* 34(2): 9-14.
1985. Nelson, J. H., R. H. Decker, A. M. Barnes, B. C. Nelson, T. J. Quan, A. R. Gillogly & G. S. Phillips. Plague surveillance using wild boar and wild carnivore sentinels. *Journal of Environmental Health* 47(6): 306-309.
1985. Campos, E. G., G. O. Maupin, A. M. Barnes & R. B. Eads. Seasonal occurrence of fleas (Siphonaptera) on rodents in a foothill habitat in Larimer County, Colorado, USA. *Journal of Medical Entomology* 22(3): 266-270.
1988. Beard, M. L., G. O. Maupin, A. M. Barnes & E. F. Marshall. Laboratory trials of cholecalciferol against *Spermophilus variegatus* (Rock squirrels), a source of human plague (*Yersinia pestis*) in the southwestern United States. *Journal of Environmental Health* 50(5): 287-289.

1988. Beard, M. L., C. E. Montman, G. O. Maupin, A. M. Barnes, R. C. Craven & E. F. Marshall. Integrated vector control. *Journal of Environmental Health* 51(2): 70-75.
1988. Barnes, A. M., T. J. Quan, M. L. Beard & G. O. Maupin. Plague in American Indians. *CDC Surveillance Summaries* 37(3): 11-16.
1989. Clover, J. R., T. D. Hofstra, B. G. Kuluris, M. T. Schroeder, B. C. Nelson, A. M. Barnes & R. G. Botzler. Serologic evidence of *Yersinia pestis* infection in small mammals and bears from a temperate rainforest of north coastal California. *Journal of Wildlife Diseases* 25(1): 52-60.
1989. Beard, M. L., C. E. Montman, G. O. Maupin, A. M. Barnes, R. B. Craven & E. F. Marshall. Field trials of the rodenticide Cholecalciferol against *Spermophilus variegatus* (Rock Squirrel) a source of human plague in the southwestern United States. *Journal of Environmental Health* 52(3): 9 pp.
1989. Thomas, R. E., M. L. Beard, T. J. Quan, L. G. Carter, A. M. Barnes & C. E. Hopla. Experimentally induced plague infection in the northern grasshopper mouse (*Onychomys leucogaster*) acquired by consumption of infected prey. *Journal of Wildlife Diseases* 25(4): 477-480.
1991. Maupin, G. O., M. L. Beard, G. Hinkson, A. M. Barnes & R. B. Craven. Studies on the control of plague in the western United States: laboratory trials of six insecticide dust formulations applied to the soil for the control of the plague vector, *Oropsylla montana* (Siphonaptera: Ceratophyllidae). *Journal of Medical Entomology* 28(6): 770-775, figs. 1-2.
1992. Beard, M. L., S. T. Rose, A. M. Barnes & J. A. Montenieri. Control of *Oropsylla hirsuta*, a plague vector, by treatment of prairie dog burrows with 0.5% Permethrin dust. *Journal of Medical Entomology* 29(1): 25-29, figs. 1-3.
1992. Beard, M. L., G. O. Maupin, R. B. Craven, C. E. Montman & A. M. Barnes. Laboratory and field trials of Permethrin-treated cotton used as nesting material to control fleas (Insecta: Siphonaptera) on cricetid rodents. *Journal of Medical Entomology* 29(2): 338-342, 1 fig.

We have also been informed of the death of **Mrs. István Szabó** (Klára), at the age of 72, after an extended illness. Friends of Dr. Szabó may send their condolences to: Dr. István Szabó, Fazekas-u. 4.I.7., H-1015 Budapest, HUNGARY.

On a happier note, **Dr. Botha de Meillon** will celebrate his 90th birthday on 15 October 1992. Cards and letters of congratulation may be sent to his residence at 8127 Cedar Road, Elkins Park, Philadelphia, PA USA 19117.

Also, Mr. F. G. A. M. Smit will celebrate his 72nd birthday on June 7th of this year. Friends may contact him at the address listed in the "Address Changes" section of this issue.

Book Reviews

Immature Insects. Volume 2. F. W. Stehr (Ed.). Kendall/Hunt Publishing Company, Dubuque, IA. 1991. ISBN 0-8403-4639-5. 975 pp. \$215.00 US.

Most entomologists, particularly systematists, are familiar with Volume 1 of this series which appeared in 1987. It was not reviewed in this newsletter since the order Siphonaptera, to which fleas belong, was not included. Volume 2 does contain an account of the larvae of this order written by Dr. Robert E. Elbel, a well-known student of flea larvae. Chapter 36, pages 674 through 689, includes a diagnosis of the order, brief comments on the life cycle and biology, a discussion of larval morphology, mounting techniques and a classification of the North American taxa based on the larvae. This is followed by a key to the families, subfamilies and tribes of the known North American flea larvae. The individual taxon accounts include references to the applicable figures, and sections on relationships and diagnosis, biology and ecology, description and comments. Each account is concluded with a selected bibliography which notes pertinent works in the bibliography at the end of the chapter. The most recent citation in the bibliography is a 1981 unpublished MS thesis, though this need not be interpreted as a criticism in view of the paucity of literature dealing with flea larvae.

The chapter serves a useful purpose in that it summarizes the current knowledge about flea larvae and introduces the reader to the terminology employed by students of this group. However, I am less than comfortable with the implication that larval morphology is useful in predicting phylogenetic affinities in the order. It is generally accepted that the larvae of holometabolous insects are free to evolve in their own directions, independent of any evolutionary trends shown by the adults, since in most cases the biological requirements of the two stages may have little or nothing in common. In this instance it is noted that

representatives of 31 of the 67 genera reported from North America have not been examined. In some cases whole tribes or even subfamilies have not been studied. The inclusion of *Mioctenopsylla* and *Conorhinopsylla* in the Hystrichopsyllidae and Leptopsyllidae, respectively, on the basis of larval characters is not credible when adult morphology is taken into account. With this reservation, the chapter provides a substantial beginning to the study of flea larvae and emphasizes the areas in need of further study.

While this is not meant to be a review of the two volumes as such, I would be remiss if attention was not drawn to a pernicious practice now rampant in the publishing industry. Namely, that of charging as much as the market will bear. At a cost of \$294.95 US plus postage, few individuals are likely to purchase these two volumes, and no teacher in possession of his or her faculties is likely to assign them as a text in their course on immature insects. In addition, the binding and paper quality are such that it is obvious that the volumes were not designed to endure the heavy usage that such works normally experience. -REL

Walker's Mammals of the World. Fifth Edition. By R. M. Nowak. The Johns Hopkins University Press, Baltimore and London. 1991. ISBN 0-8018-3970-X. Volume I, i-xlv, 1-642, plus xlix-lxii (index). Volume II, i-x, 643-1629. \$89.95 US, plus shipping.

In *Flea News* 27 (December, 1983: 101) I began a review of the Fourth Edition of this monumental work with the statement: *"Very often, the appearance of a new edition of a text involves little more than a different binding, an additional Preface, and an increase in price"*. Nothing of the kind can be said of this edition. To be sure, the price has increased, but this edition is 267 pages longer, with expanded generic accounts and a number of new illustrations.

The overall format is similar to that of the preceding edition. The front endpapers contain conversion tables for metric and US units of measurements of length, area, volume, capacity, and mass, along with a list of standard abbreviations. The conversion tables are repeated in the back endpapers, but also included are comparison of US and Metric units of weight (grams, kilograms, metric tons vs ounces, pounds and US short tons), temperature (Centigrade vs Fahrenheit) and area (hectares vs acres). A Table of Contents for both volumes is included in each volume. In volume I a short Forward by Durward Allen, Professor Emeritus of Wildlife Ecology, Purdue University, is followed by a Preface tracing the history of the preceding four editions. For the first time the Macroscelidea

(Macroscelididae) and Scandentia (Tupaiidae) are given full ordinal status. Names and content of the suborders and subfamilies are included in the ordinal and familial accounts. Subgeneric names are also included in the species lists. In all, 21 orders, 135 families, 1116 genera and 4444 species are treated. Nineteen of the genera were erected after the publication of the fourth edition. Eight generic accounts from the earlier edition have been deleted, but 106 new generic accounts have been added. A separate generic account is provided for every extinct genus known to have lived during the past 5000 years. Most of these are thought to have disappeared due to human agencies. Literature coverage extends through the end of 1988, although a few sources from 1989 and 1990 are also included.

Of particular interest to the zoogeographer is the section dealing with the world distribution of mammals. It is prefaced by a brief explanation of the terms, abbreviations and symbols employed. There follow 25 pages of listings of all of the mammalian genera and their approximate distribution. The 16 distributional categories, including the Atlantic, Indian and Pacific Oceans, do not strictly follow the Wallacean system, but they are easily translated into it.

The bulk of both volumes is taken up by the taxon accounts. Volume I treats the Monotremata, Marsupialia, Insectivora, Macroscelidea, Dermoptera, Chiroptera, Scandentia, Primates, Xenarthra, Pholidota, Lagomorpha and Rodentia-Sciuromorpha. Volume II includes the myomorph and hystrichomorph Rodentia, Cetacea, Carnivora, Pennipedia, Tubulidentata, Proboscidea, Hyracoidea, Sirenia, Perissodactyla and Artiodactyla. Each order, family and genus is described in detail. The recognized species in each genus are listed and a brief reference to the range of each is given. Generic content is not always in accord with the species lists published in Corbet and Hill (1986) or Honacki *et al.* (1982) but equivalences can usually be sorted out with a bit of detective work.

The generic accounts contain concise summaries of the taxonomic status of the group, general commentary about the appearance of its members, ecological preferences, home range, behaviour and reproductive data. Many accounts include information about the threatened or endangered status of the group, especially with regard to its economic importance and conservation.

All in all, these volumes constitute a superb exposition and it is difficult to imagine how they could be improved short of printing the photographs in color. The cost of publications has been accelerating at an alarming rate recently, and against this melancholy landscape these

volumes constitute one of the best values in publication today. -REL

MAILING LISTS

Additions

Mr. Thomas P. Bowman, Manager, Technical Services, Zoecon Corporation, 12200 Denton Drive, Dallas, TX 75234

Dr. Thomas Griffith, Department of Biology, Illinois Wesleyan University, Bloomington, IL 61702-2900.

Mr. William McLaren, P. O. Box 1288, Pebble Beach, CA 93953.

Dr. T. P. Romasheva, Sverdlova Street 1 - 12, Irkutsk, 664000, Russia

Dr. Steven A. Smith, Department of Pathobiology, Virginia-Maryland Regional College of Veterinary Medicine, VPI & SU, Blacksburg, VA 24061-0442.

Mr. David G. Stansfield, Ciba-Geigy Animal Health, P. O. Box 18300, Greensboro, NC 27419-8300.

Mr. Stephen P. Tabor, BioSystems Analysis, Inc., 4209 Lantados Street, Bakersfield, CA 93307.

Changes

Dr. Omar M. Amin, Institute of Parasitic Diseases, Diagnostic & Educational Laboratory, P. O. Box 0515, Racine, WI 53401-0515.

Dr. S. N. Rybin, 714000 - US, Osh, Kurmandjean datka 264/38, Kyrghyzstan.

Mr. F. G. A. M. Smit, 4 Glamis Gardens, Longthorpe, Peterboro, PE3 9PQ, England.

Dr. Rex Thomas, PARAVAX, Inc., 2301 Research Blvd, Suite 110, Fort Collins, CO 80526.

The following colleagues have sent reprints since **Flea News 43**:
Beaucournu, J. C., Blaski, M., Dudich, A., George, R. S., Gong Z.-d.,
Greve, J. H., Horak, I. G., McAllister, C. T., Medvedev, S. G., Rybin, S. N.,
Romasheva, T. P & Wu W.-z. For these we thank you very much!



A PREFATORY NOTE TO THE SPECIES LIST -- In keeping with the previous practice of numbering the taxa chronologically, the following list includes numbers for those taxa described in 1990. Readers are urged to call any omissions in these lists to our attention.

Addenda

Following are three taxa that have only just come to our attention or were missed in earlier numbers.

huangnanensis Ni Guo-xing, Wu Wen-zhen & Huang Gui-ping, 1984.
Amphipsylla quadratoides. [98415]. Acta biol. Plateau sin. **3**: 115-118,
from *Ochotona cansus* (= *thibetana*), PRC, Qinghai Province, Henan
Mongolzu Zizhixian, Huangnan Zangzu Zizhizhou, V.1982, HOLOTYPE
♂, QIED.

grundmanni Egoscue, 1989. **Traubella**. [98916]. Bull. Sth. Cal. Acad. Sci.
88(3): 131-134, figs. 4-7, from *Peromyscus crinitus stephensi*, USA,
Utah, Washington County, 3.2 km NE Ivins, mouth of Snow Canyon,
37.08N 113.39W, 945m, 11.XI.1988, H. J. Egoscue, HOLOTYPE ♂, USNM
No. 104628.

liui Li Chao, Wu Wen-zhen & Yang Xi-zheng, 1989. **Callopsylla**
(**Callopsylla**). [98917]. Acta zootax. sin. **14**(3): 372-375, from *Cricetulus*
kamensis, PRC, Xizang Province, Ngamring County, Rikeze, 29.48N
86.51E, 4500m, VI-VIII.1986, HOLOTYPE ♂, RIED.

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- 99001 *Hectopsylla pascuali* Beaucournu & Alcover
- 99002 *Plocopsylla consobrina* Beaucournu & Alcover

- 99003 *Agastopsylla boxi gibbosa* Beaucournu & Alcover
- 99004 *Chilropsylla allophyla tonnii* Beaucournu & Alcover
- 99005 *Geusibia digitiforma* Gong Zhen-da & Lin Jia-bing
- 99006 *Rhadinopsylla* (*Sinorhadinopsylla*) *leii* Xie Bao-qi, Gong Zhen-da
& Duan Xing-de
- 99007 *Lagaropsylla duodecima* Beaucournu & Kock
- 99008 *Ceratophyllus* (*Emmareus*) *liae* Wu Wen-zhen & Li Chao
- 99009 *Smitipsylla quadrata* Xie Bao-qi & Li Chang-jiang
- 99010 *Rowleyella nuijiangensis* Lin Jia-bing & Xie Bao-qi
- 99011 *Macrostylophora bispiniforma gongshanensis* Gong Zhen-da &
Xie Bao-qi
- 99012 *Hystriropsylla llorentei* Ayala & Morales
- 99013 *Jellisonia guerrierensis* Morales
- 99014 *Ctenoparia intermedia* Beaucournu & Kelt
- 99015 *Plocopsylla wilesi* Beaucournu & Kelt
- 99016 *Plocopsylla silewi* Beaucournu & Kelt
- 99017 *Tiamastus gallardoi* Beaucournu & Kelt
- 99018 *Liuopsylla clavula* Xie Bao-qi & Duan Xin-de
- 99019 *Xenopsylla trispinus tenuis* Beaucournu & Kock
- 99020 *Nosopsyllus wualis rongjiangensis* Li Kuei-chen & Huang
- 99021 *Nosopsyllus wualis bosensis* Li Kuei-chen & Pan
- 99022 *Nosopsyllus wualis diandongensis* Li Kuei-chen, Xie & Yang
- 99023 *Nosopsyllus elongatus longchuanensis* Li Kuei-chen, Xie & Wu
- 99024 *Nosopsyllus elongatus puerensis* Li Kuei-chen, Xie & Yu
- 99025 *Nosopsyllus elongatus yanshanensis* Li Kuei-chen, Xie & Liao
- 99026 *Coptopsylla lamellifer tarimensis* Yu, Ye & Cao
- 99027 *Vermipsylla yeae* Yu & Li
- 99028 *Stenoponia meridiana* Cao, Xie & Yu
- 99029 *Ctenophyllus* (*Ctenophyllus*) *armatus altaicus* Yu
- 99030 *Ophthalmopsylla* (*Ophthalmopsylla*) *volgensis wuqiaensis* Yu,
Shen & Ye
- 99031 *Callopsylla* (*Callopsylla*) *yui* Ye & Jiang
- 99032 *Leptopsylla algira vogeli* Beaucournu
- 99033 *Ischnopsyllus* (*Ischnopsyllus*) *kilitbahiricus* Aktas

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(Arranged in chronological order)

cuisancei Beaucournu & Morel. *Xenopsylla*. Ann. Parasit. Hum. Comp.
(1990) 65(5-6): 271-274, from *Aethomys chrysophilus*, Zimbabwe,

Kudzwe Dam, 16.56S 32.33E, 1.X.1986, D. Cuisance, HOLOTYPE ♂, MNHN, Paris.

senellarti Beaucournu & Rodhain. **Parapsyllus**. Ann. Parasit. Hum. Comp. (1990) **65**(5-6): 279-281, from *Diomedea chlororhynchos* nests, New Amsterdam Island, 37.05S 77.31E, XII.1976, J. M. Senellart. HOLOTYPE ♂, Beaucournu Collection.

poggii Beaucournu. **Xiphiopsylla**. Bull. Soc. fr. Parasit. **9**(1): 106-108, from *Heliosciurus gambianus annulatus*, Ethiopia, Shoa Province, Mahal Uonz, N Ankober, 09.30N 39.44E, 2780m, 29.I.1886, V. Ragazzi, HOLOTYPE ♂, MCSN, Genoa.

abyssinicus Beaucournu. **Ctenophthalmus (Ethioctenophthalmus) evidens**. Bull. Soc. fr. Parasit. **9**(1): 110-112, from *Arvicanthus abyssinicus*[=*niloticus*], Ethiopia, Shoa Province, Antoto, ca. 09.15N 38.50E, N Addis Ababa, 16.IX.1885, V. Ragazzi, HOLOTYPE ♂, MCSN, Genoa.

vidua Beaucournu & Gallardo. **Cleopsylla**. Bull. Soc. ent. Fr. **96**(2): 187, from *Rhyncholestes raphanurus*, Chile, Osorno Province, Lagos Puyehue, 40.42S 71.57W, 1400m, 15.IV.1988, M. Gallardo, HOLOTYPE ♀, Beaucournu Collection.

uncinatus Beaucournu & Gallardo. **Ectinorus (Ectinorus)**. Bull. Soc. ent. Fr. **96**(2): 191, from *Andinomys edax*, Chile, Parinacota Province, Belen, 18.29S 69.31W, 300m, 4.VIII.1987, M. Gallardo, HOLOTYPE ♂, Beaucournu Collection.

curvatus Beaucournu & Gallardo. **Ectinorus (Ectinorus)**. Bull. Soc. ent. Fr. **96**(2): 195, from *Phyllotis darwini*, Chile, Parinacota Province, Parinacota, 18.12S 69.16W, 4000m, 3.IX.1989, M. Gallardo, HOLOTYPE ♂, Beaucournu Collection.

plesiomorphus Beaucournu & Gallardo. **Listronius**. Bull. Soc. ent. Fr. **96**(2): 198, from *Akodon longipilis*, Chile, Malleco Province, Lonquimay, Paso Pino Hachado, 38.39S 70.54W, 1730m, 19.III.1988, M. Gallardo, HOLOTYPE ♂, Beaucournu Collection.

- singula** Rybin. **Nycteridopsylla**. Parazitologiya 25(2): 172-174, from *Barbastella leucomelas*, USSR, Kirghizia SSR, Osh Oblast, Kanigut Cave, 2000m, 18.V.1990, HOLOTYPE ♂, ZIAN, Leningrad (St. Petersburg).
- bijiangensis** Xie Bao-qi, Fang Zhu & Hu Gui. **Neopsylla biseta**. Acta zootax. sin. 16(2): 232, from *Apodemus draco*, PRC, Yunnan, Bijiang County, Bilouxue Mountain, 3000-3200m, 1978, HOLOTYPE ♂, RIED.
- jii** Xie Bao-qi & Tian Jie. **Doratopsylla**. Acta zootax. sin. 16(2): 246, from *Nasillus gracilis* = *Uropsyllus soricipes*, PRC, Yunnan, Gongshan County, Gaoligong Mountain, ca. 26.01N 98.68E, 3000-3200m, X.1985, HOLOTYPE ♂, RIED.
- liui** Xie Bao-qi & Xie Shu. **Doratopsylla**. Acta zootax. sin. 16(2): 246, from *Nasillus gracilis* = *Uropsyllus soricipes*, PRC, Yunnan, Gongshan County, Gaoligong Mountain, ca. 26.01N 98.68E, 3000-3200m, X.1985, HOLOTYPE ♂, RIED.
- jianchuanensis** Xie Bao-qi & Yang Kuang-rong. **Doratopsylla coreana**. Acta zootax. sin. 16(2): 247, from *Niviventer andersoni*, PRC, Yunnan Province, Jianchuan County, Mt Laojin, 3500m, VII-VIII.1979-1980, HOLOTYPE ♂, RIED.
- flattispina** Wu Wen-zhen, Liu Mei-li & Cai Li-yun. **Rhadinopsylla (Micropsylla)**. Acta zootax. sin. 16(3): 366-368, from *Cricetulus kamensis*, PRC, Gansu Province, Aksay Kazakzu Zizhixian, 39.04N 94.02E, 2600-2800m, no date or collector, HOLOTYPE ♂, RIED.
- ailuropodae** Jeu Ming-hua, Wang Dun-qing & Li Kuei-chen. **Chaetopsylla (Chaetopsylla)**. Sichuan J. Zool. 10(1): 9, from *Ailuropoda melanoleuca*, PRC, Sichuan Province, Pingwu County, XI.1984, HOLOTYPE ♀, ZMU.
- aiderensis** Medvedev & Alifirenko. **Phaenopsylla**. Parazitologiya 25(4): 316, from "young sandrat", Turkmenistan, Kara-Kalinskii Region, environs of Aidere, ca. 38.25N 56.46E, 40 km E Kara Kala, 11.X.1982, E. V. Drugova, HOLOTYPE ♂, ZIAN, Leningrad (St. Petersburg).

qinghaiensis Zahn Xin-ru, Wu Wen-zhen & Cai Li-yun. **Dorcadia**. Acta ent. sin. 34(4): 492-495, from *Ovis aries*, PRC, Qinghai Province, Pingan County, XII.1981, HOLOTYPE ♂, RIED.

1992

lushuiensis Gong Zheng-da & Huang Wen-li. **Ctenophthalmus** (**Sinoctenophthalmus**). Acta ent. sin. 35(1): 112, from *Eothenomys cachinus* = *Graomys griseoflavus*, PRC, Yunnan Province, Lushui County, Mt Gaoligong, 26.01N 98.68E, 2800-3200m, XI-XII.1987 & IV.1988, HOLOTYPE ♂, RIED.

xiei Gong Zheng-da & Duan Xing-de. **Ctenophthalmus** (**Sinoctenophthalmus**). Acta zootax. sin. 17(1): 99, from *Eothenomys cachinus* = *Graomys griseoflavus*, PRC, Yunnan Province, Gongshan County, Mt Dandanika, 27.69[sic!]N 98.33E, 2450-2700m, X.1987, HOLOTYPE ♂, RIED.

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LITERATURE ON SIPHONAPTERA

Although it may not be obvious from the titles, citations included here pertain to fleas and the zoonoses associated with them. Additional information is available upon written request.

(1984-1986)

1984 (List 14)

Ni Guo-xing, Wu Wen-zhen & Huang Gui-ping. A new subspecies of *Amphipsylla quadratoides* from Qinghai Province. Acta biol. Plateau sin. 3: 115-118, figs. 1-5.

1986 (List 12)

Rawlings, J. A. Lyme Disease in Texas. Zentralbl. Bakteriол. Microbiol.

(1986-1989)

Hyg., Ser. A. 263: 483-487.

Spielman, M. I., G. K. Potter, S. M. Taubman & W. R. Hodge. Pain, pruritus, and swelling localized to two toes. Archiv. Dermatol. 122(3): 329...334.

1987 (List 10)

Eigelis, Yu. K. (Ed.). The natural nidality and prophylaxis of zoonoses. Trudy protivochumn. uchrezhdh. SSSR. Saratov. 113 pp. (Contains 18 articles on plague, etc.)

Whitely, H. E., J. August, C. S. Foil, C. Griffin, L. Medleau & W. Miller. Flea-control tips from the experts. Vet. Med. 82(9): 913-916.

1988 (List 9)

Horak, I. G., J. Boomker, V. DeVos & F. T. Potgeiter. Parasites of domestic and wild animals in South Africa. XXIII. Helminth and arthropod parasites of warthogs, *Phacochoerus aethiopicus*, in the eastern Transvaal Loveld. Onderstepoort J. Vet. Res. 55(3): 145-152.

Narayana Pillai, K. G. & R. S. Prasad. Nutrition, haemolymph changes and ovarian maturation of rat fleas, *Xenopsylla cheopis* (Rothschild) and *X. astia* (Rothschild) (Insecta: Siphonaptera). Proc. Indian Nat. Sci. Acad. Part B. Biol. Sci. 54(4): 245-252.

Scarlett, J. M., N. S. Moise & J. Rayl. Feline hyperthyroidism: a descriptive and case-control study. Prev. Vet. Med. 6(4): 295-309.

Seal, S. C. Dr. Tirumurti Memorial Lecture, 1987. History of researches on plague leading to its eradication in India. Proc. Indian Nat. Sci. Acad. Part B. Biol. Sci. 54(4): 215-238.

1989 (List 7)

Caro, R. & M. Piscopo. Treatment of ectoparasitoses in fowls with cyhalothrin Spot-On. Vet. Argent. 6(51): 69-70...72-75.

(1986-1989)

Hyg., Ser. A. 263: 483-487.

Spielman, M. I., G. K. Potter, S. M. Taubman & W. R. Hodge. Pain, pruritus, and swelling localized to two toes. Archiv. Dermatol. 122(3): 329...334.

1987 (List 10)

Eigelis, Yu. K. (Ed.). The natural nidality and prophylaxis of zoonoses. Trudy protivochumn. uchrezhd. SSSR, Saratov. 113 pp. (Contains 18 articles on plague, etc.)

Whitely, H. E., J. August, C. S. Foil, C. Griffin, L. Medleau & W. Miller. Flea-control tips from the experts. Vet. Med. 82(9): 913-916.

1988 (List 9)

Horak, I. G., J. Boomker, V. DeVos & F. T. Potgeiter. Parasites of domestic and wild animals in South Africa. XXIII. Helminth and arthropod parasites of warthogs, *Phacochoerus aethiopicus*, in the eastern Transvaal Loveld. Onderstepoort J. Vet. Res. 55(3): 145-152.

Narayana Pillai, K. G. & R. S. Prasad. Nutrition, haemolymph changes and ovarian maturation of rat fleas, *Xenopsylla cheopis* (Rothschild) and *X. astia* (Rothschild) (Insecta: Siphonaptera). Proc. Indian Nat. Sci. Acad. Part B. Biol. Sci. 54(4): 245-252.

Scarlett, J. M., N. S. Moise & J. Rayl. Feline hyperthyroidism: a descriptive and case-control study. Prev. Vet. Med. 6(4): 295-309.

Seal, S. C. Dr. Tirumurti Memorial Lecture, 1987. History of researches on plague leading to its eradication in India. Proc. Indian Nat. Sci. Acad. Part B. Biol. Sci. 54(4): 215-238.

1989 (List 7)

Caro, R. & M. Piscopo. Treatment of ectoparasitoses in fowls with cyhalothrin Spot-On. Vet. Argent. 6(51): 69-70...72-75.

(1989-1990)

Clover, J. R., T. D. Hofstra, B. G. Kuluris, M. T. Schroeder, B. C. Nelson, A. M. Barnes & R. G. Botzler. Serologic evidence of *Yersinia pestis* infection in small mammals and bears from a temperate rainforest in north coastal California. J. Wildl. Dis. **25**(1): 52-60.

Coulanges, P. Plague in Tananarive (from its appearance in 1921 to its resurgence in 1979). Arch. Inst. Pasteur Madagascar **56**(1): 9-35.

Goncharov, A. I., T. P. Romasheva, B. K. Kotti, A. Bavaasan & S. Zigmed. *Opredelitel' blokh Mongol'skoi Narodnoi Respubliki.* [Keys to the fleas of the Mongolian National Republic.] Ulan Bator, 415 pp.

Haitlinger, R. Arthropod communities occurring on small mammals from non-wooded areas of the urban agglomeration of Wrocław. Acta Parasit. Polonica **34**(1): 45-66.

Hakbijl, T. Insect remains from site Q, an early Iron Age farmstead of the Assendelver Polders Project. Helinium **29**: 77-102.

Li Chao, Wu Wen-zhen & Yang Xi-zheng. Description of a new species of *Callopsylla* from Xizang, China. Acta zootax. sin. **14**(3): 372-375.

Mauri, R. & G. Navone. Some ectoparasites (Acarina and Suctoria) of Argentine marsupials. Neotropica **34**(92): 125-131 [1988].

Melo, C. R. & I. S. Melo. Verrucoid elephantoid lymphodema associated with severe infestation by *Tunga penetrans*. Anais bras. Dermatol. **64**(1): 35-37.

1990 (List 5)

Beaucournu, J. C. *Leptopsylla algira vogeli* n. ssp. (Siphonaptera: Ceratophyllidae, Leptopsyllinae), parasite de Crocidure (Mammalia: Insectivora: Crocidurinae) sur l'île de Gozo (Archipel de Malte). Bonn. zool. Beitr. **41**(3-4): 213-222, figs. 1-5.

Dudich, A. & M. Ambros. Ectoparasites of small mammals in the (1990)

(1990)

Bátorliget Nature Reserves (Acari: Mesostigmata, Ixodida[e]; Insecta: Anoplura, Siphonaptera). The Bátorliget Nature Reserves - after forty years, 1990: 675-679.

Dudich, A. & A. Stollmann. Materiály drobných zemných cicavcov (Insectivora, Rodentia) a ektoparazitov (Acarina, Anoplura, Siphonaptera) z územia Slovenskej Republiky. 3. Prehľad lokalít z východoslovenského kraja. Zborník Východoslov. Múzea Kosic. Prírod. Vedy 31: 23-40.

Dwyer, G., S. A. Levin & L. Buttell. A simulation model of the population dynamics and evolution of myxomatosis. Ecol. Monogr. 60(4): 423-447.

Calviello, R. C., E. F. Saraceno, S. A. Davidzon & G. Sanchez. Tungiasis. Arch. argent. Dermatol. 39(6): 387-391 [1989].

Difonzo, E. M., G. C. Agostini, F. Taviti & N. Petrini. Tungiasis: an infrequent consequence of overseas travel? Giorn. ital. Dermatol. Venereol. 125(11): 519-520.

Larsen, K. S. Fleas. The squirrel flea. Danish Pest Inf. Lab. Ann. Rept. 1990: 60-61.

Makundi, R. H. & B. S. Kilonzo. Activity of synergized pyrethrins against fleas, *Xenopsylla brasiliensis*, *Xenopsylla cheopis* and *Ctenocephalides felis* (Siphonaptera: Pulicidae) in the laboratory. Pyrethrum Post 18(1): 3-6.

Miller, W. H., Jr. Symmetrical truncal hair loss in cats. Compend. Cont. Ed. Pract. Veterin. 12(4): 461-470.

Olsen, A. Fleas. Danish Pest Inf. Lab. Ann. Rept. 1990: 58-60.

Romasheva, T. P. *Manual for the identification of some flea species in Siberia and the Far East.* Irkutsk, 58 pp.

Stepanov, V. M., I. L. Martinevskii, L. Koturga, S. Kh. Khamzin, G. P. Sklyarenko & A. I. Mishchenko. Epizootiological prerequisites and clinical

(1990-1991)

and epidemiological feature[s] of a case of plague in the village of Kazbek, Gur'ev district, in 1988. Zh. Mikrobiol. Epidemiol. Immunobiol. 8: 116-119.

Velimirovic, B. Plague and Glasnost. First information about human cases in the USSR in 1989 and 1990. Infection 18(6): 388-393.

Vlcek, M. & J. Máca. Succession of communities of small mammals and their ectoparasites on refuse dumps of communal waste. Univ. Carolinae Environment. 3(2): 15-37 [1989].

Wu Wen-zhen, Li Chao & Ma Yan-ning. Discovery of the male of *Paradoxopsyllus angustisinus* (Leptopsyllidae) from Qinghai, China. Entomotaxonomia 12(1): 36, 1 fig.

Yakimenko, V. V., I. I. Bogdanov & A. A. Tagil'tsev. Arthropods in rook colonies in the forest-steppe and steppe zones of west Siberia. Parazitologiya 24(6): 466-473.

1991 (List 3)

Agayeva, T. Zh. & A. D. Petrova. Population structure of the arthropods of the relict souslik's burrows in western Tian Shan. Moscow Univ. Biol. Sci. Bull. (1990) 45(3): 21-27. [English translation from Vest. Moskov. Univ., Biol. 45: 20-25.

Arends, J. J. External parasites and poultry pests. In: B. W. Calnek, J. H. Barnes, C. W. Beard, W. M. Rief & H. W. Yoder, Jr., (Eds.). *Diseases of poultry*. ISU Press, Ames, IA. pp. 702-730.

Ashford, R. W. The human parasite fauna: towards an analysis and interpretation. Ann. trop. Med. Parasit. 85(1): 189-198.

Bazanov, L. P., I. F. Zhovtyi, M. P. Maevskii, V. T. Klimov & A. F. Popkov. Seasonal dynamics of blockage formation in the flea *Citellophilus tesquorum altaicus* in the Tuva natural plague focus. Medit. Parazit. Parazitar. Bol. 1: 24-26.

Beaucournu, J. C. Deux Siphonaptères nouveaux d'Ethiopie (Xiphop-

(1991)

syllidae et Ctenophthalmidae) et essai de clé dichotomique du genre *Xiphiopsylla* Jordan et Rothschild, 1913. Bull. Soc. fr. Parasit. **9**(1): 105-118, figs. 1-9.

Beaucournu, J. C. & M. H. Gallardo. Siphonaptères du Chili; description de quatre espèces nouvelles (Siphonaptera). Bull. Soc. ent. fr. **96**(2): 185-203, figs. 1-34.

Beaucournu, J. C. & C. Guiguen. Liste commentée des puces d'Éthiopie (Insecta: Siphonaptera). Ann. Parasit. Hum. Comp. **66**(3): 126-133.

Blaski, M. Fleas (Siphonaptera) collected on the rodents from two industrial plants in Silesia. Acta Biol. Selesiana **18**(35): 84-88.

Blaski, M. A contribution to the knowledge of fleas (Siphonaptera) of Upper Silesia. The fleas caught on small mammals living on farms. Acta Biol. Selesiana **18**(35): 89-94.

Burns, D. A. A potpourri of parasites in poetry and proverb. Br. Med. J. (Clin. Res. Ed.) **303**(6817): 1611-1614.

Chadee, D. D., E. Furlonge & C. Naraynsingh. Tungiasis in Trinidad: case report. Trans. R. Assoc. trop. Med. Hyg. **85**(2): 38.

Cornack, K. M. & P. K. O'Rourke. Parasites of sheep dogs in the Charleville district, Queensland. Aust. Vet. J. **68**(4): 149.

Corpus, L. D. & K. M. Corpus. Mass flea outbreak at a child care facility: case report. Am. J. Publ. Hlth **81**(4): 497-498.

Coward, P. S. Fleas in southern England. Vet. Rec. **129**(12): 272.

Delobette, H. Myxomatosis in rabbits: a world first in vaccination. Action Vétérinaire No. 1169-1170: 19-20.

Elbel, R. Order Siphonaptera. In: F. Stehr (Ed.). *Immature Insects*. Volume 2. pp. 674-689.

(1991)

Gaede, K. Active uptake of water vapour from unsaturated atmospheres in arthropods: sorption kinetics. Zool. Jahrb. Abt. Allgem. Zool. Physiol. Tiere **95**(2): 135-171.

George, R. S. The fleas (Siphonaptera) of Vice-County 57...Derbyshire. J. Derbyshire ent. Soc., N. S. **105**: 6-11.

Heripret, D. & T. Olivry. Dermatitis due to flea bite allergy. I. Fleas and flea bite allergy. Prat. Méd. Chirur. Anim. Compag. **26**(1): 25-31.

Hinaidy, H. K. A contribution on the biology of *Dipylidium caninum*. Part II. J. Vet. Med., Series B. **38**: 329-336.

Kilonzo, B. S. The effects of permethrin on *Ctenocephalides felis*, the commonest flea pest of livestock in Tanzania. Ins. Sci. Appl. **12**(5-6): 593-598.

Kilonzo, B. S. Larvicidal effects of neem, *Azadirachta-Indica* on fleas in Tanzania. Ins. Sci. Appl. **12**(5-6): 699-702.

Lang, J. D. & W. Wills. Ecology of sylvatic plague in the San Jacinto Mountains of southern California. Bull. Soc. Vector Ecol. **16**(1): 183-199.

Maupin, G. O., M. L. Beard, G. Hinkson, A. M. Barnes & R. B. Craven. Studies on the control of plague in the western United States: laboratory trials of six insecticide dust formulations applied to soil for the control of the plague vector, *Oropsylla montana* (Siphonaptera: Ceratophyllidae). J. med. Ent. **28**(6): 770-775, figs. 1-2.

Medvedev, S. G. & V. E. Alifirenko. A new species of fleas of the genus *Phaenopsylla* (Siphonaptera: Leptopsyllidae). Parazitologiya **25**(4): 316-322, figs 1-3.

Mori, M. & F. Prignano. Murine typhus with typical cutaneous manifestations. Giorn. ital. Dermatol. Venereol. **126**(9): 445-447.

Munck, C. Control of flies and fleas on mink farms. Dansk Pelsdyravl

(1991)

54(2): 36-39.

Rao, S. S. & K. M. Rao. Insect repellent N, N-diethylphenylacetamide: an update. J. med. Ent. 28(3): 303-306.

Roberts, M. The parasites of the Polynesian rat: biogeography and origins of the New Zealand parasite fauna. Int. J. Parasit. 21(7): 785-793.

Rybin, S. N. A new flea species from south Kirghizia, *Nycteridopsylla singula* n. sp. (Siphonaptera: Ischnopsyllidae). Parazitologiya 25(2): 172-174, figs. 1-2.

Rzhevskaya, A. E., L. P. Rapoport, L. M. Orlova, K. K. Nuriev & L. P. Suslova. Fleas in the east Kisil-Kum and their epizootic significance. Parazitologiya 25(6): 504.

Teltow, G. J., P. V. Fournier & J. A. Rawlings. Isolation of *Borrelia burgdorferi* from arthropods collected in Texas. Am. J. trop. Med. Hyg. 44(5): 469-474.

Thomas, R. E. *Yersinia pestis* virulence characters and transmission events in the ecology of plague. Does *Yersinia pestis* moderate its own transmission? Bull. Soc. Vector Ecol. 16(1): 43-49.

Whitehead, M. D., H. R. Burton, P. J. Bell, J. P. Y. Arnould & D. E. Rounsevell. A further contribution on the biology of the Antarctic flea, *Glaciopsyllus antarcticus* (Siphonaptera: Ceratophyllidae). Polar Biology 11(6): 379-384.

Williams, C. K. & R. J. Moore. Inheritance of acquired immunity to myxomatosis. Aust. J. Zool. 39(3): 307-311.

Williams, E. S., E. T. Thorn, T. J. Quan & S. L. Anderson. Experimental infection of domestic ferrets (*Mustela putorius furo*) and Siberian polecats (*Mustela eversmanni*) with *Yersinia pestis*. J. Wildl. Dis. 17(3): 441-445.

Wu Wen-zhen, Liu Mei-li & Cai Li-yun. On a new species of *Rhadinopsylla* from Gansu Province, China. Acta zootax. sin. 16(3): 366-368, figs. 1-4.

(1991-1992)

Yakimenko, V. V., I. I. Bogdanov & A. A. Tagil'tsev. Arthropods of the nest complex in colonies of sand martins in west Siberia and eastern Kazakhstan. Parazitologiya 25(1): 39-41.

Zahn Xin-ru, Wu Wen-zhen & Cai Li-yun. Description of a new species of *Dorcadia* Ioff (Siphonaptera: Vermipsyllidae). Acta ent. sin. 34(4): 492-495, figs. 1-6.

1992 (List 1)

Antonelli, A. L. Flea Control. Wash. St. Coop. Ext. Svc. Bull. No. EB0817, 2 pp.

Beard, M. L., G. O. Maupin, R. B. Craven, C. E. Montman & A. M. Barnes. Laboratory and field trials of Permethrin-treated cotton used as nesting material to control fleas (Insecta: Siphonaptera) on cricetid rodents. J. med. Ent. 29(2): 338-342.

Beard, M. L., S. T. Rose, A. M. Barnes & J. A. Montenieri. Control of *Oropsylla hirsuta*, a plague vector, by treatment of prairie dog burrows with 0.5% Permethrin dust. J. med. Ent. 29(1): 25-29, figs. 1-3.

Bennett, G. F., R. A. Earle & B. L. Penzhorn. *Ornithodoros peringueyi* (Argasidae) and *Xenopsylla trispinus* (Siphonaptera), probable intermediate hosts of *Hepatozoon atticorae* in the South African cliff swallow, *Hirundo spilodera*. Can. J. Zool. 70(1): 188-190.

Cooper, J. E. Sticktight fleas (*Echidnophaga gallinacea*) on birds. Vet. Rec. 130(5): 108.

Frank, A. A., J. L. Ross & B. K. Sawvell. Toxic epidermal necrolysis associated with flea dips. Vet. Human Toxicol. 34(1): 57-60.

Ghazi, R. R. & R. Ali. Fleas (Siphonaptera) of pikas (*Ochotona rufescens*) in Baluchistan, Pakistan. Bull. Inst. Zool. Acad. sin. 31(1): 9-14.

Gong Zheng-da & Duan Xing-de. A new species of *Ctenophthalmus* from

(1992)

Yunnan Province, China (Siphonaptera: Hystrichopsyllidae). Acta zootax. sin. 17(1): 97-99, figs. 1-7.

Gong Zheng-da & Huang Wen-li. A new species of *Ctenophthalmus* from Mt Gaoligong, West Yunnan (Siphonaptera: Hystrichopsyllidae). Acta ent. sin. 35(1): 110-112, figs. 1-7.

Kern, W. H., Jr. The ectoparasites of the southeastern bat, *Myotis austroriparius*, from Florida. Bat Res. News 32(4): 77 [1991].

Kern, W. H., Jr., P. G. Koehler & R. S. Patterson. Diel patterns of cat flea (Siphonaptera: Pulicidae) egg and fecal deposition. J. med. Ent. 29(2): 203-206, figs. 1-2.

Kern, W. H., Jr., P. G. Koehler, R. S. Patterson & R. W. Wadleigh. Spectrophotometric method of quantifying adult cat flea (Siphonaptera: Pulicidae) feces. J. med. Ent. 29(2): 221-225, figs. 1-4.

Lisbôa Lopes, C. M., P. M. Linardi & J. R. Botelho. Duplication of the spermatheca in *Polygenis tripus* (Jordan) (Siphonaptera: Rhopalopsyllidae). J. med. Ent. 29(1): 111-112, figs. 1-2.

McAllister, C. T. & N. Wilson. Two additions to the flea (Siphonaptera: Hystrichopsyllidae: Leptopsyllidae) fauna of Texas. Texas J. Sci. 44(2): 245-247.

Moser, B. A., P. G. Koehler & R. S. Patterson. Effect of Methoprene and Diflubenzuron on larval development of the cat flea (Siphonaptera: Pulicidae). J. econ. Ent. 85(1): 112-116.

Richerson, J. V., J. F. Scudday & S. P. Tabor. An ectoparasite survey of mammals in Brewster County Texas, 1982-1985. SW Ent. 17(1): 7-16.

Roe, D. J. & G. D. Sales. Welfare implications of ultrasonic flea collars. Vet. Rec. 130(7): 141.

Rothschild, M. Neosomy in fleas, and the sessile life-style. J. Zool. Lond.

(1992)

226: 613-629.

Rust, M. K. Influence of photoperiod on egg production of cat fleas (Siphonaptera: Pulicidae) infesting cats. J. med. Ent. 29(2): 242-245, figs. 1-3.

Sudekum, M., R. H. Poppenga, N. Raju & W. E. Braselton. Pennyroyal oil toxicosis in a dog. JAVMA 200(6): 817-818.

Surfass, T. L., L. M. Raymon & R. P. Brooks. Ectoparasites of river otters in Pennsylvania. J. Wildl. Dis. 28(1): 138-140.

Yathiraj, S., P. M. Rao, N. R. J. Reddy, K. G. Umesh & M. T. Rai. Control of ticks and fleas on canines with Amitraz. Indian Vet. J. 69(2): 161-162.

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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

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